

Reverse-engineering NLI:

A study of the meta-inferential properties of Natural Language Inference

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What is NLI?

Task Given a pair of sentences, classify their entailment *relation* as one of:

Entailment | Contradiction | Neutral

Theoretical importance of NLI

the meaning of a sentence is that which can be inferred, under the assumption that it is true
(inferentialist semantics)

SNLI

a classic NLI dataset

- hypotheses – Flickr30k captions
- premises from human crowd workers
- 3 items for each hypothesis
- 570k items total

We will show you the caption for a photo. We will not show you the photo. Using only the caption and what you know about the world:

- Write one alternate caption that is **definitely** a **true** description of the photo. [. . .]
- Write one alternate caption that **might** be a **true** description of the photo. [. . .]
- Write one alternate caption that is **definitely** a **false** description of the photo. [. . .]

Examples (from the SNLI annotator instructions)

Hypothesis:

Two dogs are running through a field

Premise:

1. There are animals outdoors.
2. Some puppies are running to catch a stick.
3. The pets are sitting on a couch.

Label:

E

N

C

But what is it really?

Logical interpretations

	<i>Eab</i>	<i>Cab</i>	<i>Nab</i>
Material Conditional (MC)	$a \rightarrow b$	$a \rightarrow \neg b$	$\neg(a \rightarrow b) \wedge \neg(a \rightarrow \neg b)$
Strict Conditional (SC)	$\Box(a \rightarrow b)$	$\Box(a \rightarrow \neg b)$	$\Diamond(a \wedge b) \wedge \Diamond(a \wedge \neg b)$
Existential Import (EI)	$\Diamond a \wedge \Box(a \rightarrow b)$	$\Diamond a \wedge \Box(a \rightarrow \neg b)$	$(\Box \neg a \vee \Diamond(a \wedge b)) \wedge (\Box \neg a \vee \Diamond(a \wedge \neg b))$

? We'll write *Eab* to mean that sentence *a* entails sentence *b*

? $\Box/\Diamond$ universally/existentially quantify over possible (plausible) worlds (Tarskian semantics)
e.g. $\Box(a \rightarrow b)$ iff $\forall \mathcal{M}(\mathcal{M} \models a \vee \mathcal{M} \models b)$

IF the NLI labels cover the space of sentence pairs
AND all the labels (including neutral) have non-trivial content

THEN the NLI relations are not definable in propositional logic

Also: note the modal language in the SNLI and MNLI annotator instructions



For the modal logic K:

EI: the three relations partition the space of sentence pairs.

SC: the three relations cover the space (possible overlap between E and C)

Let's find out.

Main idea

We will empirically test which **meta-inferential relations** hold for models trained on a classic NLI dataset (SNLI).

From this, we can infer which **logical interpretation** of the NLI labels is learned from the original data.



These properties correspond to *symmetry of contradiction* and *transitivity of entailment*, respectively



This says that if *a* and *b* are neutral and *b* entails *c* then *c* does not contradict *a*



In these cases, the meta-inferential relations are different for the two readings of the labels *and* we can still infer something about the label of the constructed item under both readings.

Meta-inferential relations

	SC	EI
<i>Cab</i> $\rightarrow$ <i>Cba</i>	✓	✗
<i>Cab</i> $\rightarrow$ $\neg$ <i>Eba</i>	✗	✓
<i>Eab</i> $\rightarrow$ $\neg$ <i>Cba</i>	✗	✓
<i>Nab</i> $\rightarrow$ $\neg$ <i>Cba</i>	✓	✗
<i>Eab</i> $\wedge$ <i>Cac</i> $\rightarrow$ $\neg$ <i>Ebc</i>	✗	✓
<i>Eab</i> $\wedge$ <i>Cac</i> $\rightarrow$ <i>Cbc</i>	✗	✗
<i>Eab</i> $\wedge$ <i>Nac</i> $\rightarrow$ <i>Nbc</i>	✓	✓
<i>Nab</i> $\wedge$ <i>Eac</i> $\rightarrow$ $\neg$ <i>Cbc</i>	✓	✓
<i>Nab</i> $\wedge$ <i>Cac</i> $\rightarrow$ $\neg$ <i>Ebc</i>	✓	✓
<i>Eab</i> $\wedge$ <i>Ebc</i> $\rightarrow$ <i>Eac</i>	✓	✓
<i>Eab</i> $\wedge$ <i>Cbc</i> $\rightarrow$ <i>Cac</i>	✓	✓
<i>Nab</i> $\wedge$ <i>Ebc</i> $\rightarrow$ $\neg$ <i>Cac</i>	✓	✓
<i>Nab</i> $\wedge$ <i>Cbc</i> $\rightarrow$ $\neg$ <i>Eac</i>	✓	✓
<i>Cab</i> $\wedge$ <i>Nbc</i> $\rightarrow$ $\neg$ <i>Eca</i>	✓	✗
<i>Eab</i> $\wedge$ <i>Ebc</i> $\rightarrow$ $\neg$ <i>Cca</i>	✗	✓
<i>Eab</i> $\wedge$ <i>Cbc</i> $\rightarrow$ <i>Cca</i>	✓	✗
<i>Eab</i> $\wedge$ <i>Cbc</i> $\rightarrow$ $\neg$ <i>Eca</i>	✗	✓
<i>Nab</i> $\wedge$ <i>Ebc</i> $\rightarrow$ $\neg$ <i>Cca</i>	✓	✗

Generating SNLI-like items

Prompted LLMs with **similar instructions** to the original SNLI instructions + 10 few-shot examples

Provided a **hypothesis** from SNLI **as a premise** requested 3 new hypotheses (entailment, contradiction, neutral)
25k items + all 10k test set items => 75k/30k generated items

Generated datasets from 3 LLMs

Llama3.2 (3b) Llama3.3 (70b) DeepSeek-R1 (distilled; 70b)

Validating the generated data:

BERT trained on SNLI performs well on the generated items.

BERT trained on SNLI+generated items performs well on SNLI.

Small-scale human annotation shows comparable agreement on the generated items.

We construct an "inferred" test set

by recombining sentences from the original and generated test sets:

1. For each SNLI item of the form *ab* add the item *ba*
2. For each pair of SNLI items of the form *ab* / *ac*, add the item *bc*
3. For each SNLI/generated pair of the form *ab* / *bc*, add the item *ac*
4. For each SNLI/generated pair of the form *ab* / *bc*, add the item *ca*



These four rules correspond to the four sections in the tables above and to the right

What we found:

RoBERTa+SE predictions on the "inferred" test set

known relations	test item	E	C	N	SC	SC✓	EI	EI✓
<i>Cab</i>		0.4	79.5	20.1	<i>Cba</i>	79.5	$\neg$ <i>Eba</i>	99.6
<i>Eab</i>	<i>ba</i>	8.4	2.4	89.1	–	–	$\neg$ <i>Cba</i>	97.6
<i>Nab</i>		12.8	8.7	78.5	<i>Cba</i>	91.3	–	–
<i>Eab</i> $\wedge$ <i>Cac</i>		0.4	82.1	17.5	–	–	$\neg$ <i>Ebc</i>	99.6
<i>Eab</i> $\wedge$ <i>Nac</i>	<i>bc</i>	0.6	3.8	95.6	<i>Nbc</i>	95.6	<i>Nbc</i>	95.6
<i>Nab</i> $\wedge$ <i>Eac</i>		57.2	6.7	36.1	$\neg$ <i>Cbc</i>	93.3	$\neg$ <i>Cbc</i>	93.3
<i>Nab</i> $\wedge$ <i>Cac</i>		0.8	84.3	14.9	$\neg$ <i>Ebc</i>	99.2	$\neg$ <i>Ebc</i>	99.2
<i>Eab</i> $\wedge$ <i>Ebc</i>		96.8	0.8	2.4	<i>Eac</i>	96.8	<i>Eac</i>	96.8
<i>Eab</i> $\wedge$ <i>Cbc</i>	<i>ac</i>	0.7	98.3	1.0	<i>Cac</i>	98.3	<i>Cac</i>	98.3
<i>Nab</i> $\wedge$ <i>Ebc</i>		59.0	3.0	38.0	$\neg$ <i>Cac</i>	97.0	$\neg$ <i>Cac</i>	97.0
<i>Nab</i> $\wedge$ <i>Cbc</i>		0.3	87.1	12.6	$\neg$ <i>Eac</i>	99.7	$\neg$ <i>Eac</i>	99.7
<i>Cab</i> $\wedge$ <i>Nbc</i>		0.5	60.6	38.9	$\neg$ <i>Eca</i>	99.5	–	–
<i>Eab</i> $\wedge$ <i>Ebc</i>		1.3	4.4	94.3	–	–	$\neg$ <i>Cca</i>	95.6
<i>Eab</i> $\wedge$ <i>Cbc</i>	<i>ca</i>	0.1	74.3	25.5	<i>Cca</i>	74.3	$\neg$ <i>Eca</i>	99.9
<i>Nab</i> $\wedge$ <i>Ebc</i>		2.3	11.7	86.0	$\neg$ <i>Cca</i>	88.3	–	–

✦ Overall the model performs well on the meta-inference task: the model prediction is consistent with the meta-inferential prediction most of the time.

✦ Poor consistency with the meta inferences *Cab* $\rightarrow$ *Cba* and *Eab* $\wedge$ *Cbc* $\rightarrow$ *Cca* suggest the EI reading over the SC reading.

✦ Error analysis suggests some of this effect may be explained by artificial discourse phenomena like "backwards anaphora".

✦ However, relatively poor consistency with *Nab* $\wedge$ *Ebc* $\rightarrow$ $\neg$ *Cca* and *Nab* $\rightarrow$ $\neg$ *Cba* (among other results) is also suggestive of the EI reading.

Future work

Apply the method to other NLI/RTE datasets.
Is the reading of the labels consistent across datasets?

Examine the SNLI closed-world assumption.
What is pragmatically excluded by a given premise?

Test the adaptability of LLMs.
Do LLMs use different notions of inference in different contexts?

In the paper

openreview.net/forum?id=NAcvSi2CRM

Review of previous work on logical properties of NLI

Thorough investigation of valid meta-inferential relations under different modal interpretations

Experimental details, further results, and models

